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WildfireIQ Kamloops: An AI-Powered Wildfire Risk, Air Quality, and Community Preparedness Platform for the Thompson-Okanagan Region

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Abstract

Background/Objectives: Wildfires in British Columbia are growing in frequency and scale, and the Thompson-Okanagan region around Kamloops has been among the most severely affected parts of Canada over the past decade. BC's 2023 wildfire season was the largest on record, burning over 2.8 million hectares, with repeated large-scale evacuations causing serious consequences for residents, local businesses, and public health. Wildfire smoke has been linked to increased emergency room visits, school and workplace closures, and lasting respiratory harm in vulnerable populations including children, the elderly, and outdoor workers. Despite the severity and regularity of these events, the tools available to ordinary people remain scattered and incomplete. Air quality applications do not connect to fire risk data, evacuation resources are static, and no platform synthesizes available information into clear, actionable guidance. WildfireIQ Kamloops is designed to fill that gap by producing a publicly accessible web platform that brings together wildfire risk prediction, air quality forecasting, community preparedness, and 30-year climate trend analysis in one place for the Thompson-Okanagan region.

Methods: The project follows two phases. In Phase 1, all data sources are gathered from freely available government and research agencies including BC Wildfire Service, Environment Canada, NASA, BC Emergency Management, and ClimateData.ca. Two predictive models are developed: a wildfire risk model that learns from 25 years of historical fire records combined with weather conditions, fire weather index values, and satellite-measured vegetation health, producing a daily risk rating of Low, Moderate, High, or Extreme for geographic zones across the Thompson-Okanagan; and an air quality forecasting model that learns the relationship between nearby fire activity, wind conditions, and observed air quality readings in Kamloops to produce 48-hour forecasts. Both models are evaluated against held-out data from the 2022 and 2023 fire seasons to assess genuine predictive ability. In Phase 2, four platform features are built and connected to live data and model outputs: a Wildfire Risk Map with colour-coded daily risk zones and fire locations; an Air Quality Monitor showing current index, 48-hour forecast, and historical smoke events; a Community Preparedness Hub that generates tailored FireSmart checklists,

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displays active evacuation orders by neighbourhood, and tracks completed actions; and a Climate Trend module showing 30 years of fire season severity alongside projected regional temperature and precipitation changes.

Results: This project is in progress. The anticipated outcomes are a fully operational, publicly accessible platform with all four features live and documented; two trained predictive models with complete methodology descriptions and accuracy results reported transparently and reproducibly; and a final research report published to TRUSpace alongside a fully open codebase available to other researchers, students, and community organizations who wish to extend or adapt the work. The wildfire risk model is expected to produce meaningful daily risk differentiation across geographic zones based on the richness of 25 years of historical records, and the air quality forecasting model is expected to deliver reliable 48-hour smoke condition forecasts based on the established relationship between fire proximity, wind direction, and observed readings in Kamloops.

Conclusions: WildfireIQ Kamloops addresses all four of TRU's sustainability pillars. Environmentally, it produces original research linking historical fire patterns, daily weather, and satellite data to forward-looking risk assessments for this specific region. Socially, it delivers a practical tool for TRU students and Kamloops residents who face real risk each summer. Economically, it models how smoke exposure drives healthcare demand, which is relevant to regional planning decisions. Culturally, the platform is built to incorporate Traditional Ecological Knowledge on fire management as those datasets become available through the Wildfire Resilience Consortium of Canada, of which TRU is a foundational partner. Findings will be shared through TRUSpace, a public research event presentation, and proactive engagement with the TRU Sustainability Office, and the full codebase will be released as open source for others to build on.

Keywords: wildfire risk prediction; air quality forecasting; community preparedness; Thompson-Okanagan; machine learning; environmental informatics; smoke forecasting; FireSmart; climate trends; public health